Public contract: JSON surfaces and exit codes

This document declares the stability status of every machine-readable surface BackupSage exposes, and how the golden fixtures that freeze them work. The compatibility window (how long these promises hold, across which versions) is tracked separately in issue #57.

Golden fixtures

Every surface below is frozen as a committed fixture under `tests/fixtures/contract/` and compared on every test run by `tests/contract.rs` — ten fixtures: one per surface plus completed-with-skips variants of `search --all --json` and `dedup --json` (populated `skipped[]` / `archives_offline`), plus `dedup_chain.json`, which pins the keeper-star fields (#9) at non-degenerate values — a real transitive-only member with `actionable: false` and `hamming_to_keep > 3` — plus two more non-degenerate content-mode corpora (#71): `search_only.json` (a `search-only` archive's `mode` field on `search --all --json`) and `metadata_only_skips.json` (a `metadata-only` archive's worded skip on `dedup --json`, alongside a normal duplicate pair so `summary.groups` stays non-zero too):

```
cargo test --test contract          # verify against the frozen contract
BACKUPSAGE_BLESS=1 cargo test --test contract   # regenerate deliberately
```

Comparison is semantic JSON equality after normalizing the only run-varying values (temp-directory prefixes become `<TMP>`, the wall-clock `indexed_unix` becomes `"<TS>"`). Any field rename, removal, addition, or type change fails the suite. **The contract is additive-only**: a re-bless whose diff adds fields may ship; a re-bless whose diff renames or removes fields is a breaking change and must not. Fixture generation only reads archives and writes to fresh temp paths — BackupSage never rewrites or deletes from an archive.

JSON surfaces

Surface	Shape	Status
`dedup --json`	Typed structs in `src/report.rs`, top-level `"version": 1`	**Stable.** Versioned, additive-only, consumed by scripts and the future web UI. v1.0.2 (#9) added `Member.actionable`, `Group.review_only_bytes`, `Summary.transitive_only_files`/`review_only_bytes`, `params.actionable_rule` — additive, still version 1. `reclaimable_bytes`/`duplicate_files` now count only keeper-star-safe members (correctness fix: previously inflated by transitive-only members whose distance to the keeper exceeds the threshold). v1.0.2 (#71) also added `ReportArchive.content_mode` and a `metadata-only` case in `summary.skipped_archives` (alongside the existing `v2-limited` case) — additive, still version 1. A metadata-only-only master now exits 2 where it previously exited 0 with an empty, unexplained report.
`search --json`	`{query, hits[], truncated, mode}`; hits carry `path`, `matches`, `snippet`, and `path_bytes` (hex) only for non-UTF-8 paths	**Frozen-as-observed.** No version field yet; fixture-protected; changes must be additive. v1.0.2 (#70) added top-level `mode` (`full`/`search-only`/`metadata-only`) — additive. On a `search-only` index `matches` is `null`: contentless FTS cannot run `highlight()`, and `snippet` is absent for the same reason. Full-mode output is byte-identical to earlier v1.x.
`search --all --json`	`{archives[{archive, truncated, mode, hits[]}], skipped[{archive, reason}]}`	**Frozen-as-observed.** Same rules as `search --json`, including `matches: null` from a `search-only` child. A `metadata-only` child is never searched: it appears in `skipped` with a worded reason, which is exit 2. v1.0.2 (#71) added the per-archive `mode` field — additive, present on every successfully-searched archive. `--snippets` against a search-only archive now also prints a stderr note, matching the single-archive path's existing note.
`master list --json`	Array of `{archive_id, label, source, source_type, db_path, schema_version, files, completed, status, indexed_unix, content_mode}`	**Frozen-as-observed.** Same rules. v1.0.2 (#71) added `content_mode` — additive.
`master verify --json`	Array of `{archive_id, label, status, source}`	**Frozen-as-observed.** Same rules.

"Frozen-as-observed" means: the shape carries no version field today, but the golden fixtures pin it exactly, so any non-additive change is caught in CI. Giving these surfaces a version field is itself an additive change and may happen in a later release.

Known residual gaps (fields the fixtures pin only at a degenerate value, so a type change to them would not be caught): `hardlink_of` is null on every corpus member (no hardlink entries yet), and `archives_incomplete` appears only as an empty array. `sparse` stays pinned false in dedup output by design — sparse rows are excluded from dedup candidates at the SQL level — so its evidence lives elsewhere: the #64 differential corpus (`tests/fixtures/sparse/`, four GNU-tar-built dialect archives, frozen-as-observed) proves old-GNU logical size/bytes/BLAKE3 against GNU tar across tar/gz/zstd, pins the PAX name-only shape, and drives every malformed-map case to a loud abort in `tests/sparse.rs`. The element shape of `summary.skipped_archives` is additionally pinned by an inline assertion in `tests/cli.rs` (v2-limited flow). Field renames and removals of all of these are still caught — the keys themselves are in the fixtures.

All other commands (`index`, `top`, `inspect`, `master add/sync/rm`) produce human-oriented text only — **unstable**, no machine-readable promise.

Exit codes

Documented in `src/main.rs` and executed as a per-subcommand matrix in `tests/contract.rs::exit_code_matrix`:

Code	Meaning
`0`	Completed cleanly. Includes "no results" and "no duplicates".
`1`	Error: bad input, missing file/index/master, unknown key or path.
`2`	Completed **with skips**: `dedup` skipped offline/incomplete/v2-limited archives; `search --all` skipped offline or unreadable archives (v2-limited archives are fully searchable and do not cause exit 2); or `master verify` found any non-`ok` archive.

Caveat frozen as observed behavior: **clap usage errors also exit 2** (unknown flag, missing subcommand). They are distinguishable from completed-with-skips — usage errors print help/usage text to stderr and perform no work. Scripts distinguishing the two should treat exit 2 with a usage message on stderr as an invocation bug, not a skip report.